Sukhun Ko

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Summary

M.S. student in CMLab at Chung-Ang University, supervised by Prof. Jihyong Oh. My research focuses on diffusion-based image compositing and real-time controllable video generation with flow matching, which I am extending toward interactive world models. I also work on low-level vision, including implicit neural representations, and 3D reconstruction.

Work Experience

Generative AI Intern, ConnectBrick Sep. 2024 – Dec. 2024
Contributed to FoodStyler, a **generative AI-based image compositing** project under Jector AI.

Project Leader, Research Project with DEXTER STUDIOS Sep. 2025 – Dec. 2025
Led an **image-to-image style transfer** pipeline for user photo stylization (commissioned project "Back-end Platform Development for Mobile Application Development," funded by DEXTER STUDIOS).

Research Intern, UCLA Art|Sci Center Spring 2026
Participated in the Research Internship Program as part of an **international research collaboration** between art and science.

Education

2025 – Feb. 2027 (expected) M.S. in Imaging Science, AI Content, Chung-Ang University, Seoul (GPA: 4.34/4.5)
2021 – 2025 B.S. in Big Data Convergence, Sangmyung University, Seoul (GPA: 4.0/4.5)

Publications

Sukhun Ko*, Seungho Choi*, Soo Ye Kim†, and Jihyong Oh†. "StreamCompose: Controllable Streaming Video Compositing." *In preparation.* Collab. w/ **Adobe Research**.

1. **Sukhun Ko**, Soo Ye Kim†, and Jihyong Oh†. "Chameleon: Style-Content Disentangled Framework for Cross-Domain Object Compositing." *Under Review.* Collab. w/ **Adobe Research**.
2. **Sukhun Ko**, SeokHyun Youn, Dahyeon Kye, Kyle Min, Chanho Eom, and Jihyong Oh†. "FLAIR: Frequency- and Locality-Aware Implicit Neural Representations." *CVPR 2026 Findings.* Collab. w/ **Oracle**.
3. **Sukhun Ko***, Seokhyun Youn*, Weeyoung Kwon, and Jihyong Oh†. "Band-Localized Splatting: Frequency-Selective Appearance Modeling for High-Frequency Surface Reconstruction." *ICIP 2026 Workshop* (Oral).
4. Dahyeon Kye, Changhyun Roh, **Sukhun Ko**, Chanho Eom, and Jihyong Oh†. "AceVFI: A Comprehensive Survey of Advances in Video Frame Interpolation." *IEEE TCSVT*, 2026.
5. **Sukhun Ko**, Chanho Eom, and Jihyong Oh†. "Implicit Neural Representations: A Holistic Survey of Techniques, Applications, and Challenges." *IEIE SPC*, 2026.

* Equal contribution. † Corresponding author.

Research Interests

Generative AI	World Models, Controllable Video Generation, Compositing, Style Transfer
Low-Level & 3D Vision	3DGS, NeRF, Implicit Neural Representations, Frame Interpolation

Skills

Generative Models	Flow-matching DiTs (FLUX.1-dev, Wan2.1), Diffusers, FSDP training (8×H100)
Frameworks & Tools	PyTorch, OpenCV, Docker, Gradio, TensorFlow
Programming	Python, C/C++, Linux, AWS

Last updated: October 31, 2026